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HIGHER-ORDER LOGIC

INTRODUCTION

What is nowadays the central part of any introduction to logic, and indeed to some the logical theory par excellence, used to be a modest fragment of the more ambitious language employed in the logicist program of Frege and Russell. 'Elementary' or 'first-order', or 'predicate logic' only became a recognized stable base for logical theory by 1930, when its interesting and fruitful meta-properties had become clear, such as completeness, compactness and Löwenheim-Skolem. Richer higher-order and type theories receded into the background, to such an extent that the (re-) discovery of useful and interesting extensions and variations upon first-order logic came as a surprise to many logicians in the sixties.

In this chapter, we shall first take a general look at first-order logic, its properties, limitations, and possible extensions, in the perspective of so-called 'abstract model theory'. Some characterizations of this basic system are found in the process, due to Lindström, Keisler-Shelah and Fraïssé. Then, we go on to consider the original mother theory, of which first-order logic was the elementary part, starting from second-order logic and arriving at Russell's theory of finite types. As will be observed repeatedly, a border has been crossed here with the domain of set theory; and we proceed, as Quine has warned us again and again, at our own peril. Nevertheless, first-order logic has a vengeance. In the end, it turns out that higher-order logic can be viewed from an elementary perspective again, and we shall derive various insights from the resulting semantics.

Before pushing off, however, we have a final remark about possible pretensions of what is to follow. Unlike first-order logic and some of its less baroque extensions, second and higher-order logic have no coherent well-established theory; the existent material consisting merely of scattered remarks quite diverse with respect to character and origin. As the time available for the present enterprise was rather limited (to say the least) the authors do not therefore make any claims as to complete coverage of the relevant literature.

1 FIRST-ORDER LOGIC AND ITS EXTENSIONS

The starting point of the present story lies somewhere within Hodges' (this volume). We will review some of the peculiarities of first-order logic, in order to set the stage for higher-order logics.

D.M. Gabbay and F. Guenthner (eds.),
Handbook of Philosophical Logic, 2nd Edition, Volume 1, 189–243.

1.1 Limits of Expressive Power

In addition to its primitives *all* and *some*, a first-order predicate language with identity can also express such quantifiers as *precisely one, all but two, at most three*, etcetera, referring to specific finite quantities. What is lacking, however, is the general mathematical concept of *finiteness*.

EXAMPLE. The notion 'finiteness of the domain' is not definable by means of any first-order sentence, or set of such sentences.

It will be recalled that the relevant refutation turned on the *compactness theorem* for first-order logic, which implies that sentences with arbitrarily large finite models will also have infinite ones.

Another striking omission, this time from the perspective of natural language, is that of common quantifiers, such as *most, least*, not to speak of *many* or *few*.

EXAMPLE. The notion 'most A are B' is not definable in a first-order logic with identity having, at least, unary predicate constants A, B. This time, a refutation involves both compactness and the (downward) *Löwenheim–Skolem theorem*: Consider any proposed definition $\mu(A, B)$ together with the infinite set of assertions 'at least n A are B', 'at least n A are not B' ($n = 1, 2, 3, \ldots$). Any finite subset of this collection is satisfiable in some finite domain with $A - B$ large enough and $A \cap B$ a little larger. By compactness then, the whole collection has a model with infinite $A \cap B$, $A - B$. But now, the Löwenheim–Skolem theorem gives a countably infinite such model, which makes the latter two sets equinumerous — and 'most' A are no longer B: in spite of $\mu(A, B)$.

One peculiarity of this argument is its lifting the meaning of colloquial 'most' to the infinite case. The use of infinite models is indeed vital in the coming sections. Only in Section 1.4.3 shall we consider the purely *finite* case: little regarded in mathematically-oriented model theory, but rather interesting for the semantics of natural language.

In a sense, these expressive limits of first-order logic show up more dramatically in a slightly different perspective. A given *theory* in a first-order language may possess various 'non-standard models', not originally intended. For instance, by compactness, Peano Arithmetic has non-Archimedean models featuring infinite natural numbers. And by Löwenheim–Skolem, Zermelo-Fraenkel set theory has countable models (if consistent), a phenomenon known as 'Skolem's Paradox'. Conversely, a given model may not be defined categorically by its complete first-order theory, as is in fact known for al (infinite) mathematical standard structures such as integers, rationals or reals. (These two observations are sides of the same coin, of course.) Weakness or strength carry no moral connotations in logic, however, as one may turn into the other. Non-standard models for analysis